

Anonymization: VPN, Tor, Proxies & OPSEC

Why this matters (legitimately)

Threat researchers, journalists, and defenders under attack use anonymity. You should understand the *model* so you can advise, not because you need to hide.

The anonymity toolbox

Tool	Model	Strength	Weakness
VPN	You → trusted provider's server → site	One hop from your IP	Provider sees everything; traffic correlation
Tor	3 random relays, layered encryption	No single hop knows both sides	Slow; exit-node sniffing; some sites block
Proxy	Single hop (HTTP/SOCKS)	Simple	Provider logs; weak
Proxychains	Multi-hop proxy chain	Obscures path	Slows; the chain knows hops

Tor circuit:

```
[ Client ] -> (Guard relay) -> (Middle relay) -> (Exit relay) -> [Target]
onion-encrypted segments visible only to each hop
```

The leaks that break anonymity

DNS requests sent outside Tor/VPN

Browser fingerprinting (canvas, plugins, webRTC IP)

Account associations (you log into GMail 'anonymously')

Metadata (file times, EXIF, your real timezone from fonts/timing)

Correlation attacks (timing + traffic size linking entry and exit)

OPSEC beats everything

The tools do *not* make you anonymous; a disciplined workflow does:

One identity per purpose; never cross accounts

No personal info anywhere in the persona's artefacts

DNS + browser + plugins consistent on the anonymized channel

Understand logs: the guards keep your start/end times

When the law intersects

Anonymizing to commit crimes is still a crime. Being anonymous is legal; evading detection *softens nothing* – where laws bind, they bind regardless of network path.

[← Honeypots & Deception](#)

[Anti-Forensics & Log Clearing →](#)